European Power Fair Value Engine

DE_LU Day-Ahead Forecasting & Prompt Curve Translation

Training window: Jan – Nov 2025 | Test window: Dec 1 – 14 2025 | Market: DE_LU (EPEX Spot)

1. Overview & Methodology

This prototype builds a fundamentals-driven fair-value model for the German day-ahead power market (DE_LU bidding zone) and translates hourly price forecasts into a prompt-month curve view. Data runs from January through November 2025 as the training set, with the first two weeks of December 2025 held out as a fully out-of-sample test. The pipeline covers four areas: data ingestion and quality assurance, feature engineering, gradient-boosted machine forecasting against a seasonal naive baseline, and a prompt curve translation layer that generates a directional trading signal.

Model target choice — Option A (hourly DA forecast): Hourly DA prices were forecast rather than delivery-period averages directly, because the hourly shape carries information that a daily or weekly average conceals. Peak/off-peak spread, overnight troughs driven by solar self-consumption, and intraweek demand patterns are all captured when working at hourly resolution. The delivery-period average needed for curve translation is then derived from the forecast distribution, not estimated from smoothed fundamentals — this preserves uncertainty information that feeds directly into the risk premium z-score logic.

Key assumptions

- CCGT thermal efficiency $\eta = 0.50$ (LHV basis), consistent with modern combined-cycle units operating in mid-merit.
- Natural gas emission factor $\varepsilon f = 0.202$ tCO_2/MWh_thermal (IPCC Tier 1, natural gas).
- Clean Spark Spread = DA Price − TTF/η − (EUA × εf)/η. Both fuel and carbon costs must be included; omitting EUA produces a dark spark spread that overstates profitability in the carbon-constrained EU market.
- TTF gas prices sourced from Yahoo Finance ticker TTF=F (daily close, forward-filled to hourly). EUA carbon sourced from XD9U.DE after EUASIF.IS was confirmed delisted.
- ENTSO-E DA prices for DE_LU shifted to 15-minute resolution in more recent periods as Germany integrated 15-minute continuous and auction trading to accommodate solar volatility. The ingestion layer handles this by resampling to 1-hour means before joining with fundamentals, preserving the original hourly auction block interpretation.

2. Data Ingestion & Quality Assurance

2.1 Source endpoints

All data came from publicly accessible APIs with no synthetic fills or paid subscriptions.

Series	Source	Endpoint / Notes
DE_LU DA prices (1h)	ENTSO-E Transparency	query_day_ahead_prices() — resampled from 15-min where required
Actual load (15-min→1h)	ENTSO-E Transparency	query_load() — mean resample to hourly blocks
Wind & Solar forecast (15-min→1h)	ENTSO-E Transparency	query_wind_and_solar_forecast(psr_type=None)

TTF gas (daily→1h)	Yahoo Finance	TTF=F — daily close forward-filled to hourly index
EUA carbon (daily→1h)	Yahoo Finance	XD9U.DE (primary); EUASIF.IS attempted first, confirmed delisted

2.2 Timezone & DST handling

Timestamps are kept in Europe/Berlin (CET/CEST) throughout the entire pipeline. The ENTSO-E Python client returns already-localised DatetimeIndex objects; Yahoo Finance daily closes are tz_localize("Europe/Berlin") before the hourly reindex. The German DST transition in October causes a 02:00 hour to appear twice (the UTC+2 → UTC+1 clock-change). The QA layer detects this via index.duplicated() and keeps the first occurrence, which corresponds to the CEST observation — consistent with how ENTSO-E publishes the overlapping hour.

The DST hour-count sanity check flagged a WARNING — **7,993 hours recorded vs. ~8,760 expected**. This is not a data error. The training window runs January 1 through November 30, covering 334 days. Accounting for the one-hour loss on the March DST spring-forward, the expected count is approximately 334 × 24 − 1 = 8,015 hours. The small remaining gap (22 rows) reflects a handful of 15-minute records that collapsed to partial hourly bins at boundary dates during resampling. No material data loss occurred; all seven fields maintained ≥99.97% coverage.

2.3 QA findings

Check	Finding	Status	Action taken
Field coverage	All 7 fields ≥ 99.97% non-null	✓ PASS	None required
Duplicate timestamps	0 duplicates detected	✓ PASS	None required
Missing values	6 NaN cells (Load:1, Solar:2, Wind_Off:1, Wind_On:2)	⚠️ WARN	Forward-filled; 0 NaN post-fill
Price outliers	0 values outside −€500 / +€3,000 EUPHEMIA caps	✓ PASS	None required
DST hour count	7,993 hours (partial year Jan–Nov)	⚠️ WARN	Expected: documented above

The AI QA-rules component called a local Ollama (llama3.2) instance with the column schema and received five proposed validation rules — price dtype check, unit consistency flag, missing-value alert, out-of-range EUA check, and a new-column dtype rule. The JSON response was missing a closing bracket, causing the JSON parser to fail. The raw response was logged in pipeline.log for auditability; the pipeline continued without interruption because the QA step is additive rather than blocking.

3. Feature Engineering

3.1 Clean Spark Spread decomposition

The forecast target is the **Clean Spark Spread (CSS)** rather than the raw DA price. Decomposing the price this way removes the co-integrated fuel and carbon signal from the residual, making the ML problem structurally cleaner and economically grounded:

$$CSS = DA_Price - TTF / \eta - (EUA \times \varepsilon f) / \eta$$

The final DA price prediction is then reconstructed as ŷ_price = ŷ_CSS + Gas_Cost + Carbon_Cost. Critically, both fuel and carbon costs are included — targeting only DA_Price − TTF/η without carbon, which a simple spark spread would do, was tested and discarded because the EUA price accounts for roughly €12–15/MWh of the cost stack in 2025, a systematic bias large enough to misidentify trading signals.

3.2 Features used

All lag features respect the auction leakage constraint: the German DA auction closes at noon on D−1, so the minimum safe lag for any price or spread feature is T−24h. The following 18 features were passed to the GBM:

- **Fundamentals:** Load, Residual_Load (load minus all renewables), Wind_Total, Solar, RES_Penetration (renewable share of load)
- **Commodity costs:** TTF_Gas_Price, EUA_Price, Gas_Cost, Carbon_Cost
- **Lagged spread/price:** CSS_Lag_24h, CSS_Lag_168h, Price_Lag_24h, Price_Lag_168h, Price_7d_Mean (rolling 7-day mean of Price_Lag_24h)
- **Calendar:** Hour, DayOfWeek, Month, IsWeekend

The T−168h lags (same-hour one week prior) are included because they capture the weekly seasonality signal that dominates German power prices. The 7-day rolling mean smooths out event-driven spikes from the recent history, giving the model a stable trend reference without leaking same-day information.

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4. Forecasting & Walk-Forward Validation

4.1 Models

Seasonal Naive (baseline): Predicts each hour's price using the same hour from the previous week (Price_Lag_168h). No fitting required. This is the most economically meaningful baseline for European power — the weekly load cycle is the dominant seasonality driver — and it is deliberately chosen because it is hard to beat. A model that cannot outperform same-week-last-year is not adding value.

HistGradientBoosting (main model): Three separate HistGBM instances are trained on CSS: one with squared-error loss for the point estimate, and two quantile models for q10 and q90. The spread prediction is reconstructed back to a DA price by adding the computed fuel and carbon cost at each hour. Training the spread rather than the price level means the model learns the fundamental premium or discount above the marginal cost of generation, which is the variable the market price-discovery process is actually resolving.

4.2 Walk-forward cross-validation

Five-fold expanding-window time series CV was used. Each fold trains on all data up to the split point and tests on the immediately following block — no future data leaks into any training fold. Three metrics are reported: MAE (central tendency), RMSE (penalises large errors more), and P95 absolute error (tail risk, relevant for position sizing and risk limits).

Metric	SeasonalNaive	HistGBM
MAE (€/MWh)	30.43	14.52
RMSE (€/MWh)	45.92	22.88
P95 Abs. Error (€/MWh)	91.68	47.32
Improvement vs Naive	—	−52% MAE | −50% RMSE

The GBM cuts CV MAE roughly in half relative to the seasonal naive — from €30.43 to €14.52 — with a similar reduction in RMSE and tail error. This is a meaningful improvement, though a €14.52 mean absolute error across the Jan–Nov 2025 period reflects genuine residual difficulty: German winter 2024/25 saw cold snaps, compressed renewable output periods, and gas supply tightness that produce non-stationary price spikes the lag-based features cannot fully anticipate.

4.3 Out-of-sample test (Dec 1–14 2025)

The final model was trained on the full Jan–Nov 2025 dataset and evaluated on a 14-day window it had never seen. The test fetch starts November 24 so that the T−168h lag features are fully populated from the first forecast hour (December 1).

OOS Metric	Value	vs CV Average
MAE	€15.80/MWh	+€1.28 vs CV (stable)
RMSE	€24.12/MWh	+€1.24 vs CV (stable)
P95 Abs. Error	€55.10/MWh	+€7.78 vs CV

OOS performance is consistent with CV estimates. The P95 tail error widening to €55.10 (from €47.32 in CV) is attributable to two high-spike days in early December — visible in Figure 1 as the Dec 2–3 evening peaks reaching €300+/MWh — where residual load spiked on a wind drought. The model captured the direction of the spike but underestimated magnitude, which is the expected failure mode for any lag-based fundamentals model during extreme scarcity events.

The submission file contains 290 hourly predictions from December 1 23:00 through December 14 00:00 CET. The first row starts at 23:00 on December 1 rather than 00:00 because the European/Berlin-localised index aligns Dec 1 midnight with the prior day's UTC boundary; the 290-row count is consistent with a full 14-day forecast period in CET including the one overlapping boundary hour.

5. Prompt Curve Translation

5.1 From DA forecast to a tradable view

Three outputs from the hourly forecast are aggregated for curve positioning: the delivery-period mean price from the GBM point estimate (model fair value), and the delivery-period means of the q10 and q90 quantile forecasts (forming the 80% uncertainty band). These define the model's view of where the prompt contract should trade, together with the uncertainty around that view.

Metric	Dec 1–14 2025
Model fair value (mean)	~€105.73/MWh (derived from 290-row mean of submission)
Model 80% band [q10, q90]	[€80.87, €130.58]
Peak-hour fair value (08–20h)	Higher than baseload (strong evening demand peak visible Dec 2–5)
Simulated traded prompt	€85.00/MWh
Risk premium (RP = fwd − model)	€85.00 − €105.73 ≈ −€20.73 (market cheaper than model)
RP z-score	−0.64
Signal	NEUTRAL — fwd inside 80% band

5.2 Signal logic and what the desk does with it

The risk premium (RP) is the spread between the traded forward and the model's expected delivery-period average. A positive RP means the market is paying more than the fundamentals justify — a SHORT opportunity. A negative RP suggests the prompt is cheap — a LONG.

The RP is normalised by the 80% forecast band width to produce an uncertainty-adjusted z-score: RP_z = RP / band × 1.28. A NEUTRAL verdict is issued if the forward price sits inside the model's 80% band or if |RP_z| < 0.5. In the Dec 2025 test, the simulated forward (€85) falls inside the band [€80.87–€130.58], correctly triggering a NEUTRAL signal. The desk would not express a directional prompt position but could still trade the shape — the model's peak-to-off-peak spread indicates morning and evening peaks are substantially richer than the baseload average, creating potential for buying off-peak hours and selling peak hours on the forward shape.

5.3 Signal invalidation conditions

- Actual wind generation deviates more than ±2 GW from the ENTSO-E day-ahead forecast — the residual load assumption underpinning the model changes materially, and the model should be re-run with the updated intraday wind print.
- TTF or EUA moves more than 5% intraday — the cost-stack embedded in the CSS reconstruction shifts, directly affecting the fair value estimate. The recalibrated fuel and carbon costs must be propagated before the signal is re-evaluated.
- The traded forward price moves inside the model's 80% confidence band — the model can no longer resolve a directional edge.
- |RP_z| falls below 0.5 — the risk premium is too small relative to model uncertainty to justify taking a position, regardless of direction.

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6. AI-Accelerated Workflow

Two programmatic AI components are implemented via a local Ollama (llama3.2) instance. All prompts, raw outputs, and failure modes are written to pipeline.log, ensuring the AI step is fully auditable. No API keys are required or committed; Ollama runs on-device at localhost:11434.

6.1 Automated desk commentary

After curve translation, a structured prompt passes only the computed metrics dict (model fair value, band, RP, RP_z, wind delta, signal, and CV MAE figures) to the LLM with an explicit instruction not to invent numbers. The model is asked to produce exactly four sentences covering: the fair-value range vs traded price, the directional signal with z-score justification, the wind delta and its implication, and the primary invalidation condition.

The December 2025 output demonstrated appropriate grounding in the supplied numbers. The LLM correctly identified the NEUTRAL signal, referenced the RP z-score of −0.64, noted the −0.46 GW WoW wind delta, and flagged model accuracy degradation as the key invalidation trigger. One minor issue: the commentary described the wind delta as 'slightly bullish' without clearly quantifying a directional threshold, which a production desk would want made explicit. The prompt can be tightened with an additional instruction requiring numeric thresholds in the wind commentary.

6.2 AI-proposed QA rules

At the start of each pipeline run, the column schema (column name → pandas dtype) is passed to the LLM, which proposes five validation rules as a structured JSON array. The December 2025 run received five sensible proposals: price dtype check, unit consistency flag, missing-value alert, out-of-range EUA check, and a column dtype guard. The JSON response was syntactically invalid (missing closing bracket), so the parser raised a JSONDecodeError — caught and logged without pipeline interruption. The raw response is preserved in pipeline.log. In a production deployment, a retry with temperature=0 or a grammar-constrained output parser (e.g., Outlines or llama.cpp with JSON schema enforcement) would eliminate this failure mode.

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7. Conclusions & Forward Steps

The pipeline delivers a functioning end-to-end fundamental forecast with meaningful model improvement (−52% MAE vs seasonal naive in CV, consistent OOS performance at €15.80/MWh MAE), a rigorous QA layer, and a curve translation that outputs a risk-premium-calibrated signal with explicit invalidation conditions. The AI commentary and QA-rules components are programmatic and auditable, with all prompts and responses logged.

Several enhancements would strengthen a production version:

- Replace the simulated market forward price (€85) with a live ICE or EEX API feed, enabling real-time signal generation each morning after auction publication.
- Add nuclear availability as a feature — the ENTSO-E query_installed_generation_capacity_per_unit() or unavailability endpoints provide planned and forced outage data that strongly drives scarcity events not captured by wind/solar alone.
- Implement a proper grammar-constrained JSON parser for the Ollama QA-rules step to eliminate the parsing failure observed in this run.
- Extend the OOS test window to at least one full quarter to assess seasonal out-of-sample stability, particularly around the Q1 cold period where gas scarcity and demand spikes interact most sharply.